

Challenges to Modus Ponens

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1. Recap

2. Apparent counterexamples to Modus Ponens

A counterexamples to Modus Ponens would be a case where we have three sentences A, B, C, such that A and B are true, C is not true, and A is 'If B, C' (or 'If B, then C').

- Given a Kratzer-style syntax a question immediately arises about what the final condition means, given that we want to think of sentences not as mere strings of letters but as structured entities. But let's barge ahead for now and see how this plays out.

Orthodoxy says there are no counterexamples to Modus Ponens. But here are some seeming candidates.

- (i) The paradox of gentle murder (Forrester): A: 'If John kills his mother, he ought to kill her gently'. B: 'John will kill his mother'. C: 'John ought to kill his mother gently'.
- (ii) (A slight variant:) A 'If we're going to leave tomorrow we should start packing now'; B: 'We're going to leave tomorrow'; C: 'We should start packing now'.
- (iii) (McFarlane and Kolodny): A: 'If the miners are in shaft 1 we should flood shaft 2'; B: 'The miners are in shaft 1'; C: 'We should flood shaft 2'.
- (iv) (Yalcin): A: 'If the marble is big, it's likely red'; B: 'The marble is big'; C: 'The marble is likely red'.
- (v) (A slight variant): A: 'If it's Tuesday we must be in Paris'; B: 'It's Tuesday'; C: 'We must be in Paris'.
- (vi) (A different kind of variant): A: 'If the marble is big, I'm confident that it's red'; B: 'The marble is big'; C: 'I'm confident that the marble is red'. [Also consider versions with 'know' and 'think'.]
- (vii) (McGee): A: 'If a Republican wins, then if Reagan doesn't win, Anderson will win'; B: 'A Republican will win'; C: 'If Reagan doesn't win, Anderson will win'.
- (viii) (Better): We don't know yet who won. A: 'If a Republican won, then if Reagan didn't win, Anderson won'; B: 'A Republican won'; C: 'If Reagan didn't win, Anderson won'.

3. Possible reactions to these counterexamples

(Note that one could react differently to different examples.)

- (a) We're just making a mistake when we judge that A is true in these examples, or when we judge that C is false.
- (b) So much the worse for Modus Ponens! Challenge: why does it work when it works?
- (c) Give up on the idea that conditionals can be true or false. Challenge: is there some other way of making sense of 'valid' in such a setting? Does Modus Ponens come out 'valid' in the new sense, or not?
- (d) "Syntactic mismatch": the trios aren't really of the relevant syntactic forms at all—i.e. despite appearances, B and C do not occur as *constituents* in A (at least on the natural parsing of A).
1. *First version*: wide-scoping. E.g.: the logical form of 'If it's Tuesday we must be in Paris' is 'Must [[if it's Tuesday] [we be in Paris]]', or maybe 'We must [[be in Paris] [if it's Tuesday]]'. (Or something involving 'movement' which gets the same results in a somewhat more principled way.)
 - A very old idea—the medievals had a distinction between *necessitas consequentis* and *necessitas consequentiae* which was basically this. Often combined with an account on which the embedded conditional is material. But this is not compulsory.
 2. *Second version*: Kratzerian syntax. E.g. the logical form of 'If it's Tuesday we must be in Paris' is '[Must [if it's Tuesday]] [we be in Paris]'.
- (e) "Semantic mismatch": certain key words that appear both in A and C—e.g. 'should', 'likely', the inner 'if' in the McGee example—are ambiguous (or context-sensitive), and are not being interpreted in the same way when we judge A true and C false. If we fix the interpretation of these words in C to be the same one that's natural for A, then C is true.